

Experiment No. 9

Date: / /

Aim: To determine heat of displacement (enthalpy change) of 'Cu' from CuSO_4 solution by zinc dust.

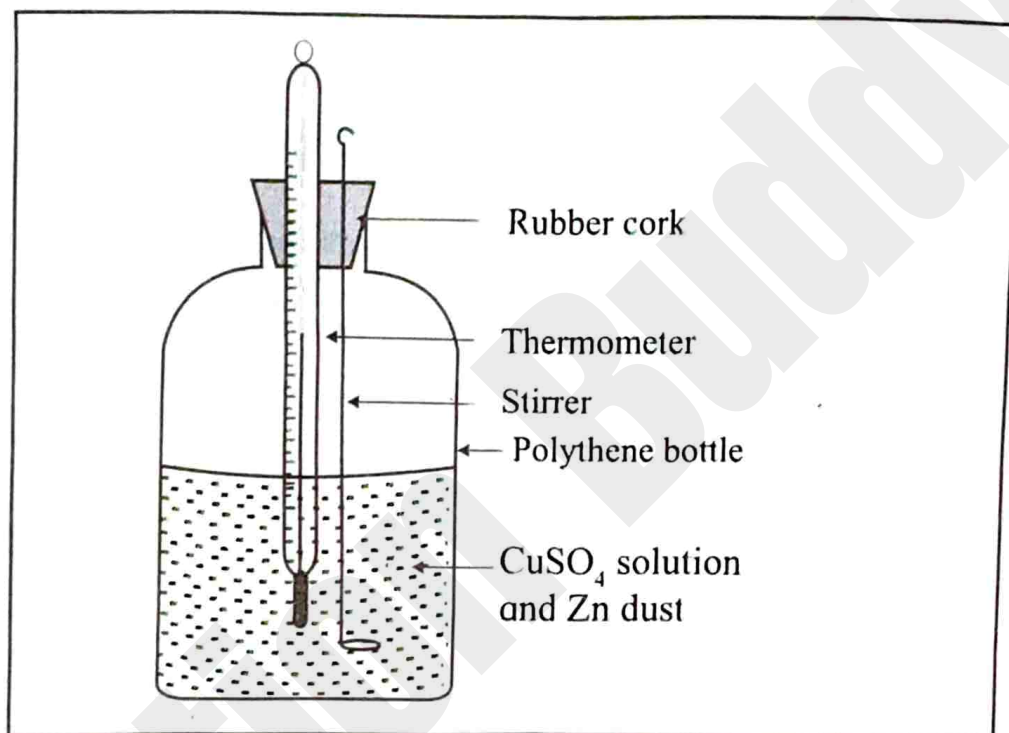
Theory: The elements having lower values of standard potential can displace another element from its solution having higher value of standard potential. The enthalpy change that occurs, when one mole of a substance is completely displaced from its aqueous solution is called enthalpy of displacement. Zinc dissolve in the copper sulphate solution by liberating heat and copper gets precipitated.

Reaction: $\text{CuSO}_4(\text{aq}) + \text{Zn}(\text{s}) \longrightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s}) \downarrow$; $\Delta H = -217.5 \text{ kJmol}^{-1}$

Apparatus: Polythene bottle, cork with two holes, beakers, thermometer ($1/10^\circ$), measuring cylinder, etc.

Chemicals: Zinc dust, 0.2 M copper sulphate solution.

Diagram:



Procedure:

1. Take 25mL 0.2 M copper sulphate solution in a clean and dry polythene bottle.
2. Record its initial temperature ($t_1^\circ\text{C}$).
3. Take 1 g of zinc dust (powder), add it to the CuSO_4 solution in polythene bottle and shake well.
4. Record the maximum (final) temperature of the solution ($t_2^\circ\text{C}$).

Note :

- i. The shaking of the solution till the blue colour of the CuSO_4 solution becomes colourless and copper metal gets deposited.
- ii. Specific heat capacity of commonly used PET type of polythene bottle can be approximated to $1 \text{ J/g } ^\circ\text{C} = 1000 \text{ J/kg } ^\circ\text{C} \approx 0.25 \text{ cal/g } ^\circ\text{C}$
- iii. If the bottle is thin walled, its mass can be neglected if mass of water inside it is more than 100g.
- iv. If a heavy bottle of different material is used, we need to know and use its water equivalent

Observation Table:

1	Mass of 0.2 M CuSO_4 solution	-	25 g
2	Mass of zinc metal	-	1 g
3	Total mass of solution	m	26 g
4	Initial temperature of CuSO_4 solution	t_1	25 °C
5	Final temperature of mixture	t_2	36 °C
6	Rise in temperature	$\Delta t = 36^\circ\text{C} (t_2) - 25^\circ\text{C} (t_1)$	11 °C
7	Specific heat capacity of water	S	4.184 J/g °C

Calculation:

Heat evolved in displacement reaction is given by

$$Q = m S \Delta t$$

$$Q = 26 \times 4.184 \text{ J/g } ^\circ\text{C} \times 11 \Delta t$$

$$Q = 1196.624 \text{ J}$$

1000 mL 0.2 M CuSO_4 solution contain 0.2 mol of 'Cu'

25 mL 0.2 M CuSO_4 solution contain

$$= \frac{25 \times 0.2}{1000}$$

$$= 5 \times 10^{-3} \text{ mol of 'Cu'}$$

..... (Q) is the heat evolved, when 5×10^{-3} mol of 'Cu' is displaced it

Therefore enthalpy of displacement is

$$\Delta H = \frac{\text{.....}(Q)}{5 \times 10^{-3}} = 239.32 \times 10^3$$

$$\Delta H = -239.32 \times 10^3 \text{ J/mol}$$

$$\Delta H = -239.32 \text{ kJ/mol}$$

Space for log calculation

Number	log
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Result:

The enthalpy of displacement of 'Cu' from copper sulphate by zinc dust $\Delta H = \text{----- kJ/mol}$

(Negative sign is because reaction is exothermic in nature)

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. The number of water molecules present in hydrated CuSO_4 is/are

a. 1

c. 3

b. 2

☒ d. 5
2. The metal used for connection of railway tracks after a particular distance

a. Zn

c. K

☒ b. Al

d. Fe
3. A thermodynamic state function is a quantity

a. used to determine heat changes

c. used to determine pressure volume work

☒ b. whose value is independent of path

d. whose value depends on temperature only
4. The enthalpies of all elements in their standard state are

a. unity

c. < 0

☒ b. zero

d. different for each element

Short answer questions

1. Why does Zinc displace copper from CuSO_4 solution?

Ans. Zn displaces copper from CuSO_4 solution because zinc is having large value for standard oxidation potential ($E_{\text{ox}} = +0.763 \text{ V}$) than copper ($E_{\text{ox}} = -0.337 \text{ V}$).

2. What is enthalpy of displacement?

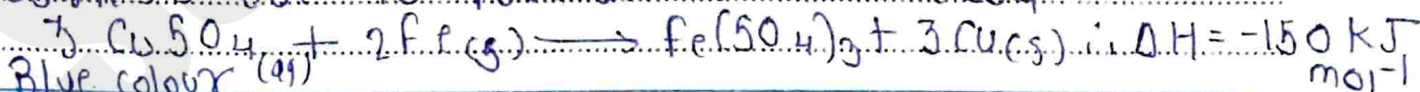
Ans. The enthalpy change that occurs when one mole of substance is completely displaced from its aqueous solution is called enthalpy of displacement.

3. What is specific heat of a substance?

Ans. The specific heat of substance i.e. the amount of energy required to raise the temp. of 1gm of the substance by 1°C unit $\text{J/g}^\circ\text{C}$ or $\text{cal/g}^\circ\text{C}$.

4. Write balanced chemical equation, when Iron filings are kept in blue coloured solution of copper sulphate.

Ans. When Iron filings are kept in blue coloured solution of copper sulphate then blue colour slowly becomes colourless due to formation of FeSO_4 .



Remark and sign of teacher: _____